

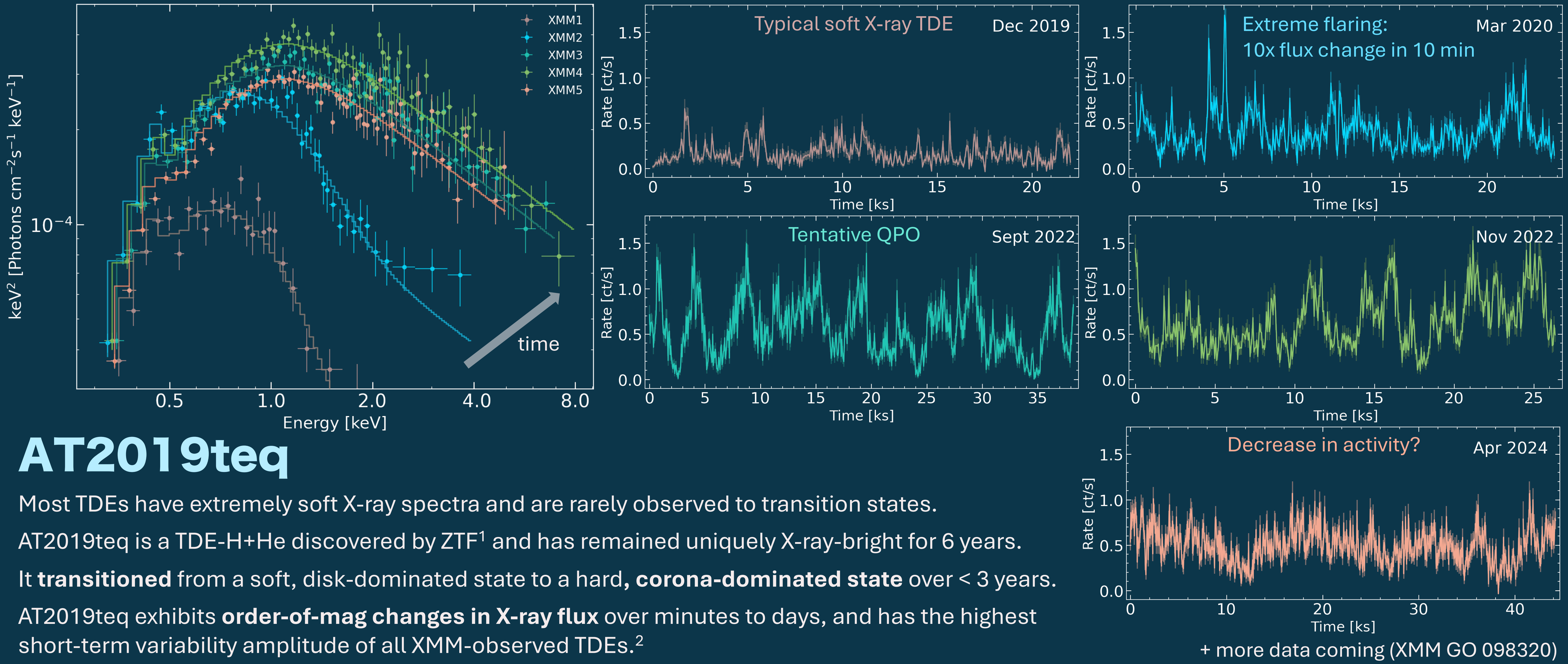
Disk-to-corona State Transition and Extreme X-ray Variability in a Tidal Disruption Event

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Paper here:

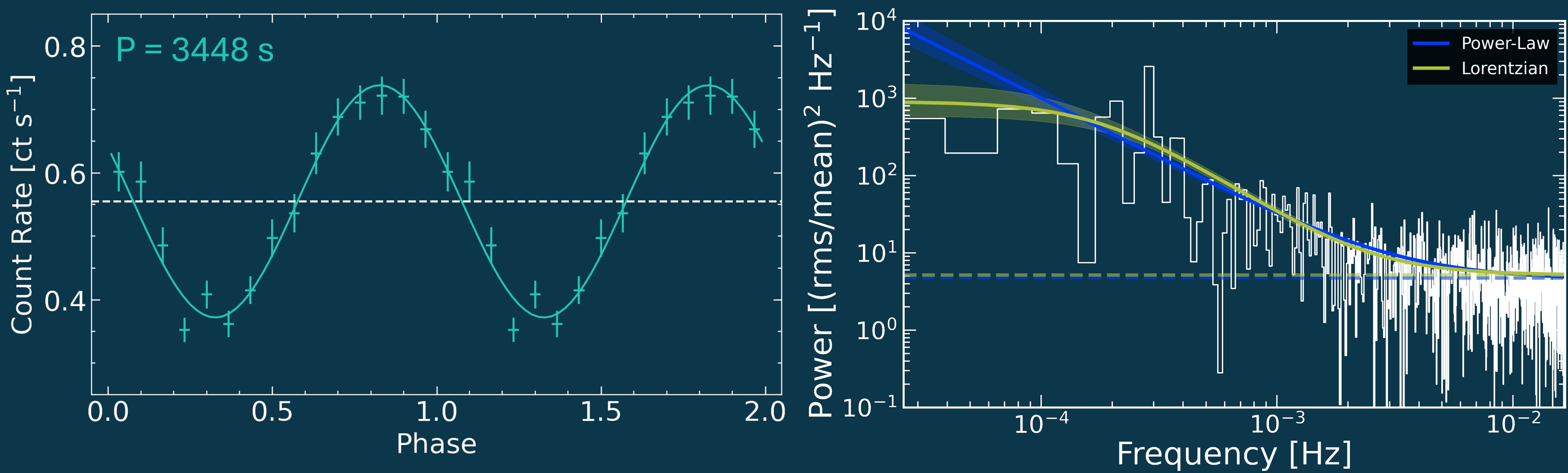


- ☆ **Do tidal disruption events (TDEs) have accretion state transitions like XRBs?**
- ☆ **Can we measure black hole masses well enough to use TDEs as probes of accretion physics?**



AT2019teq

Most TDEs have extremely soft X-ray spectra and are rarely observed to transition states. AT2019teq is a TDE-H+He discovered by ZTF¹ and has remained uniquely X-ray-bright for 6 years. It **transitioned** from a soft, disk-dominated state to a hard, **corona-dominated state** over < 3 years. AT2019teq exhibits **order-of-mag changes in X-ray flux** over minutes to days, and has the highest short-term variability amplitude of all XMM-observed TDEs.²

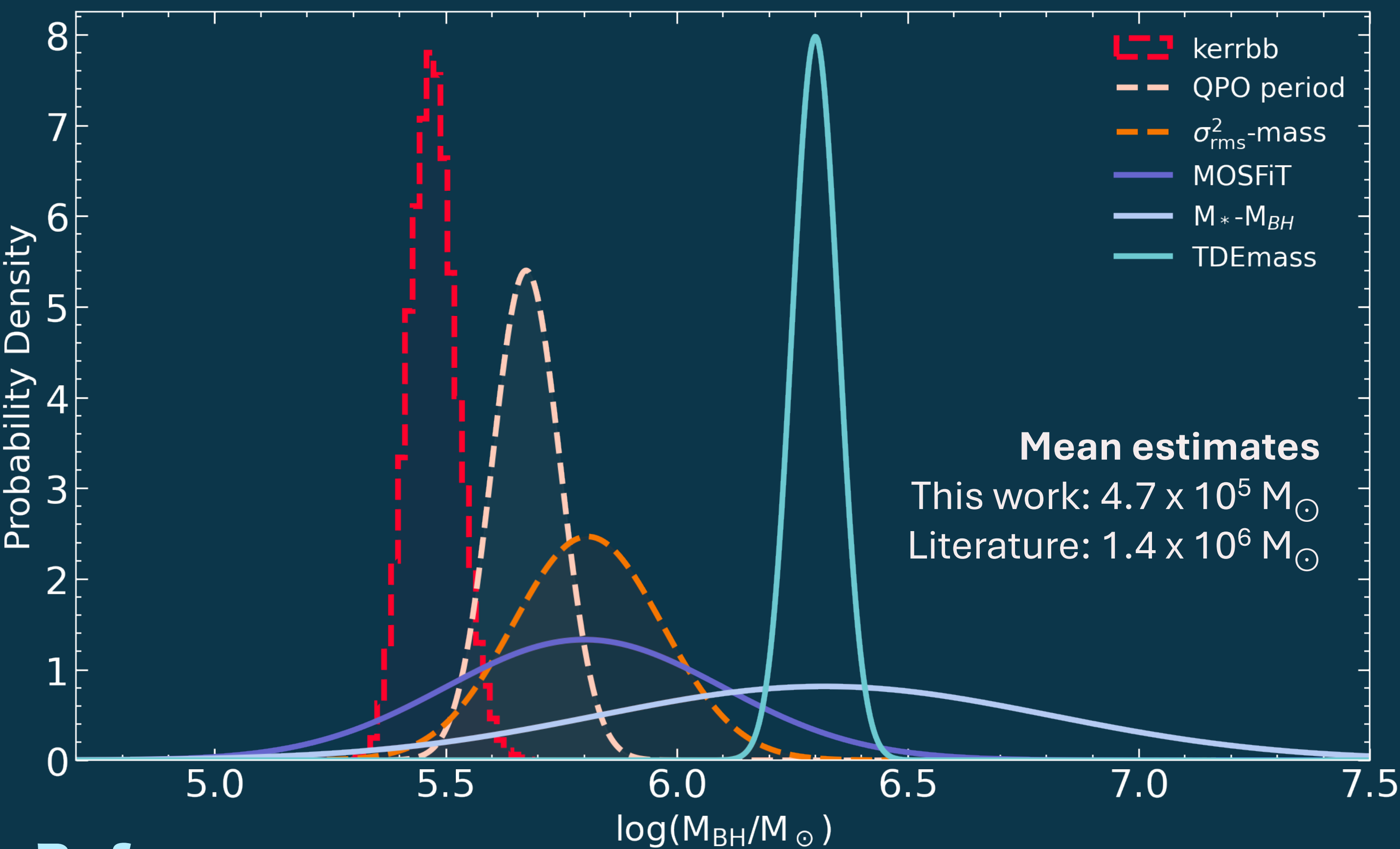
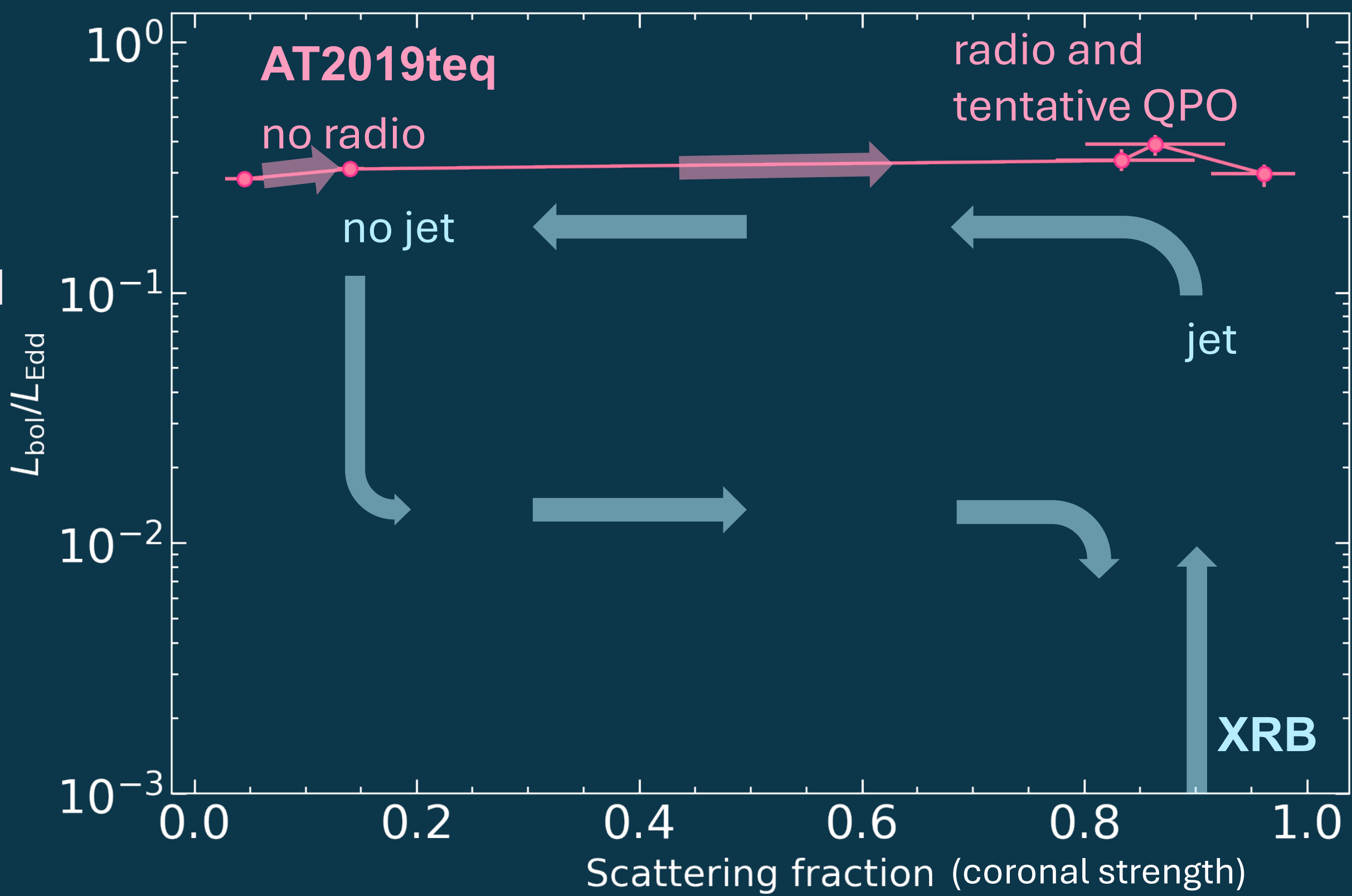


Tentative sub-mHz QPO

The power spectrum in one epoch shows a sharp spike **>10x above the continuum** at 0.29 mHz ($P = 3448$ s). We tested significance via red noise simulations ($p < 0.03$) and AIC model selection ($p < 0.007$). p-values are model-dependent and the feature is transient: we consider this a marginal detection.

Connections to XRB outbursts

Canonically, outbursting X-ray binaries (XRBs) follow a q-shaped trajectory through the hardness-intensity diagram, forming a compact jet and low-frequency QPOs in the hard state. AT2019teq shows some parallels to XRB outbursts: it exhibits a **late-time radio brightening**³ and a tentative QPO around the onset of the hard state. However, its state transition over < 3 yr corresponds to minutes–hours on $10 M_{\odot}$ XRB timescales: **>100x faster** than expected from mass scaling, and at higher luminosity.



Black hole mass from X-rays

We harnessed AT2019teq's soft/hard accretion states to apply X-ray-based mass estimates from both regimes, mutually consistent within 2.2σ :

- Soft state: **disk modelling** with kerrbb⁴
- Hard state: **AGN mass-variability** relation⁵
- **QPO frequency**: associate period with Keplerian motion at the ISCO

The mean is systematically lower, but consistent to 2.3σ , with estimates from UV/optical light curve modeling and host galaxy scaling relations.^{1,6}

This puts AT2019teq on the **low-mass** end of both TDE and MBH populations and shows promise of X-ray-based BH mass estimates for TDEs.

References

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[5] Ponti G. et al. 2012 A&A **542** A83
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